LPS003 - Hurricane Isaias

October 12, 2026

1 Observations of The Regulus Project

Focus: Historical Audit: 2026-10-9 05:00:00 to 2026-10-11 0:59:59

Min Pressure 1003.0 mb
Max Pressure 1016.98 mb
Max Wind Speed 51.5 mph
Total Rainfall 2.77 in (70.35 mm)

1.1 Executive Summary

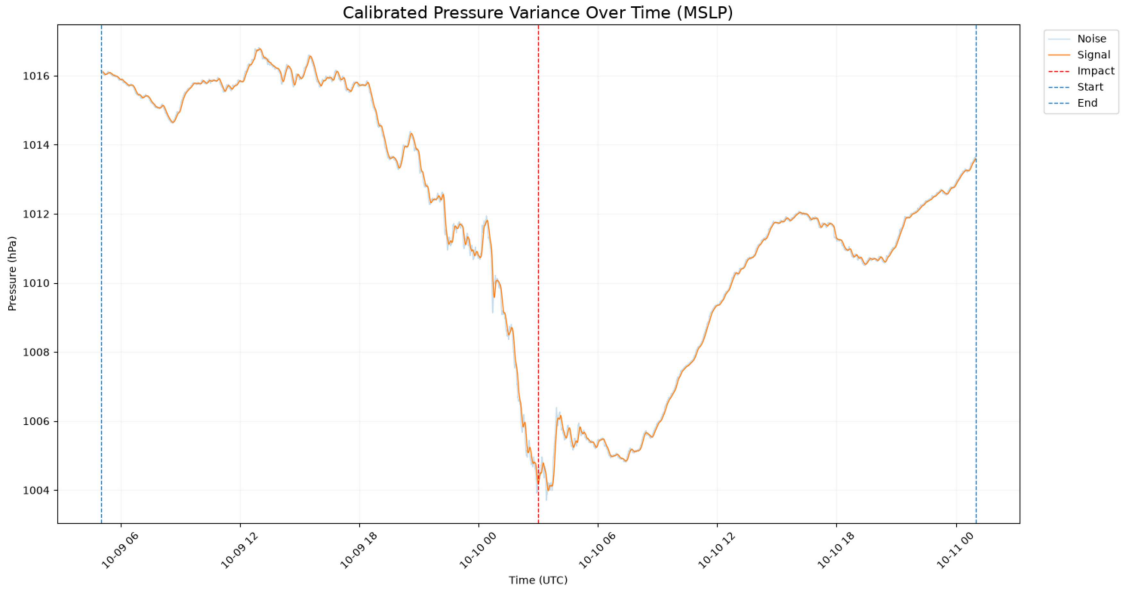
Low Pressure System 003 (Hurricane Isaias) impacted the facilities envelope in a variety of ways. Primarily with excess winds and rain.

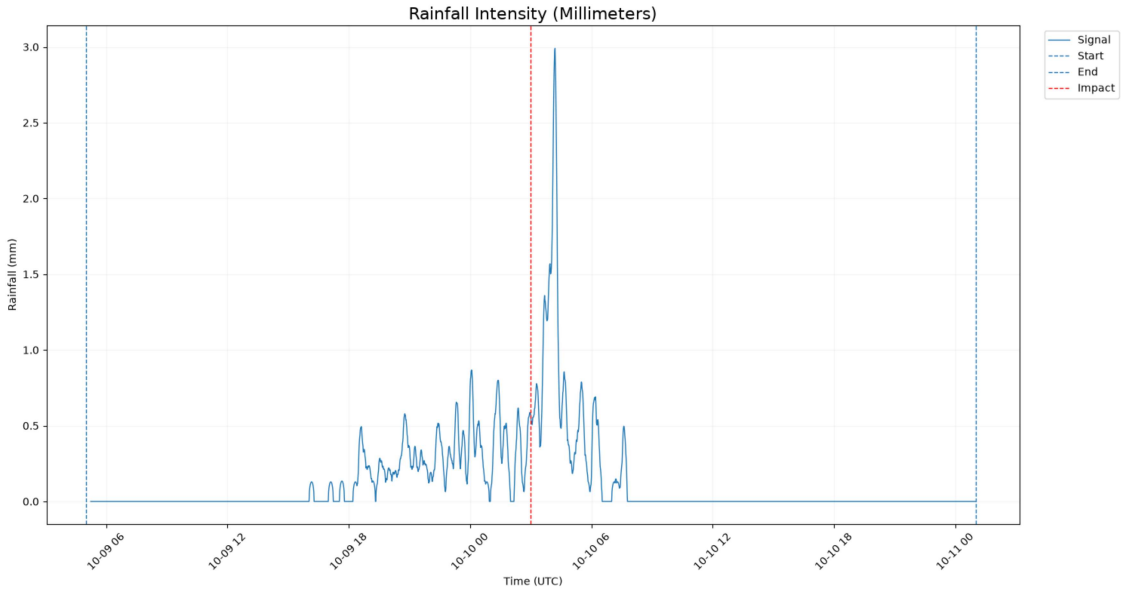
1.2 Meteorological Context

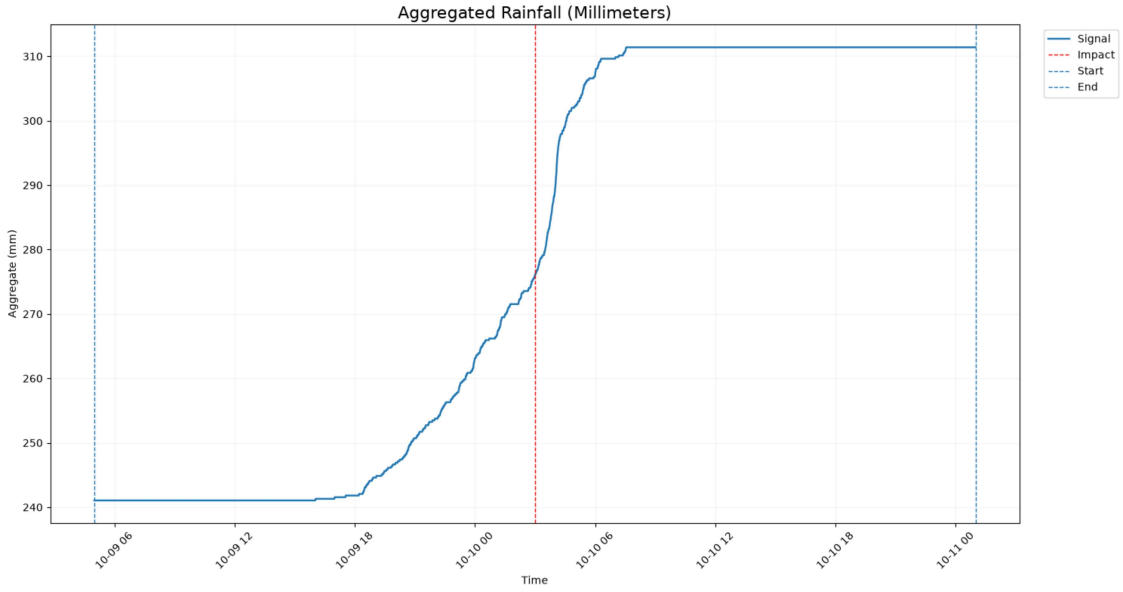
This system began impacting the air column surrounding the facility on 10-09-2026 and the first rainfall hit within the same day. The surrounding towns roughly 25+ miles away saw significantly more damage and wind. The path of the storm was close enough that the system was able to caputre some valuable insights.

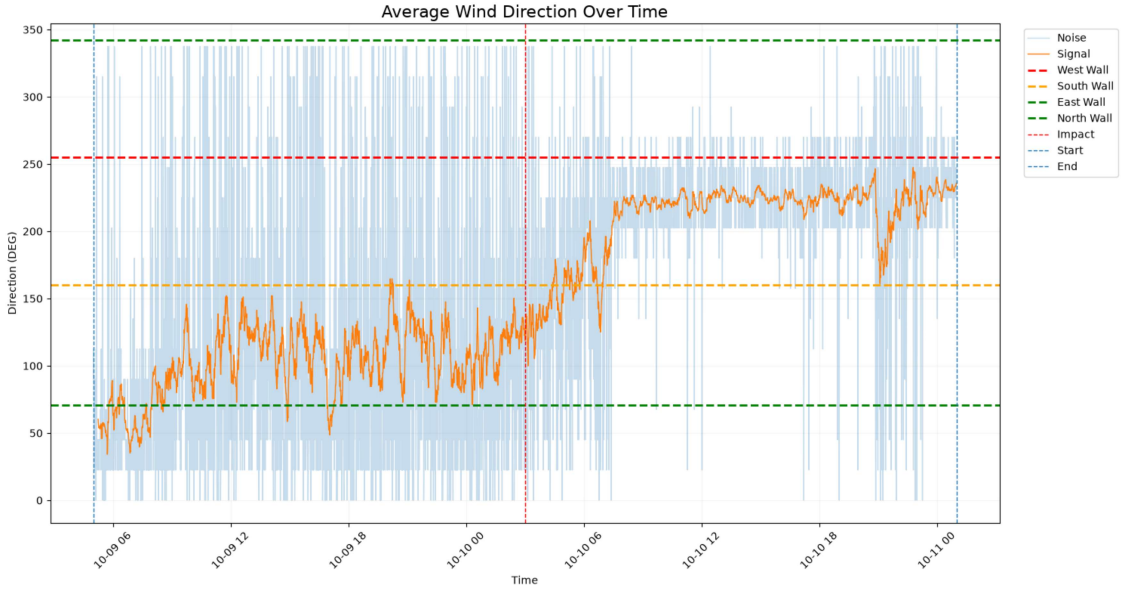
1.3 Infrastructure Statement

Our property lost internet access on 10/09/2026 and access has not been restored as of 10/12/26. Power infrastrucrture integrity waivered significantly on 10/09 and 10/10. With a UPS and second generation stations these infrastructure outages did not impact the observations of the system or interfere with the scientists ability to generate the reports.

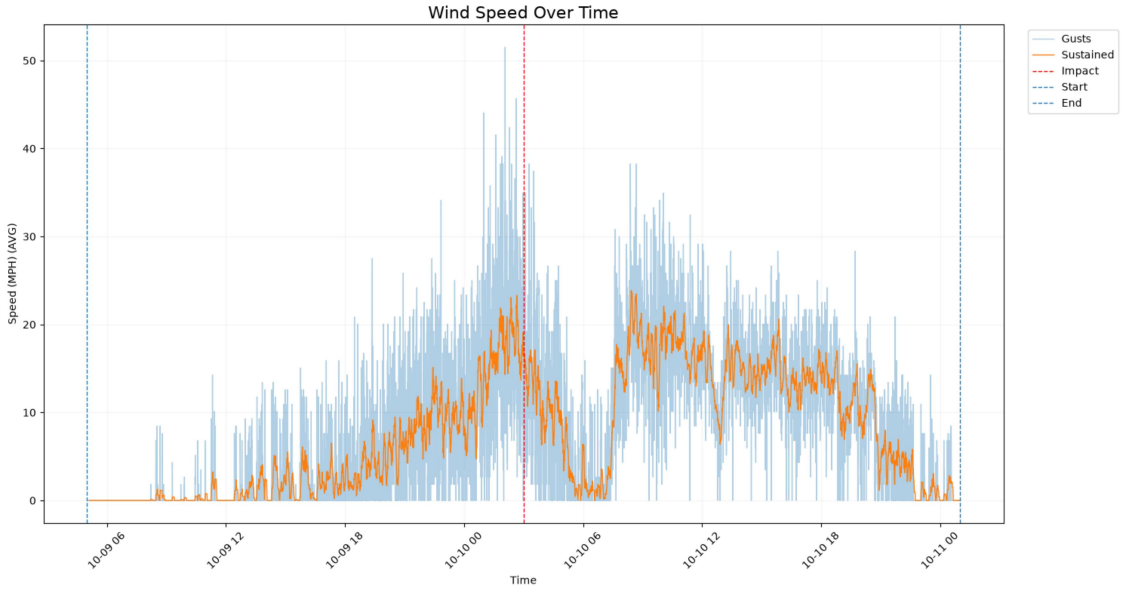

Calibrated Pressure Variance Over Time (MSLP)
Noise
Signal
Impact
Start
End
Pressure (hPa)
1016
1014
1012
1010
1008
1006
1004
10-09 06
10-09 12
10-09 18
10-10 00
10-10 06
10-10 12
10-10 18
10-11 00
Time (UTC)


Rainfall Intensity (Millimeters)
Signal
Start
End
Impact
Rainfall (mm)
3.0
2.5
2.0
1.5
1.0
0.5
0.0
10-09 06
10-09 12
10-09 18
10-10 00
10-10 06
10-10 12
10-10 18
10-11 00
Time (UTC)





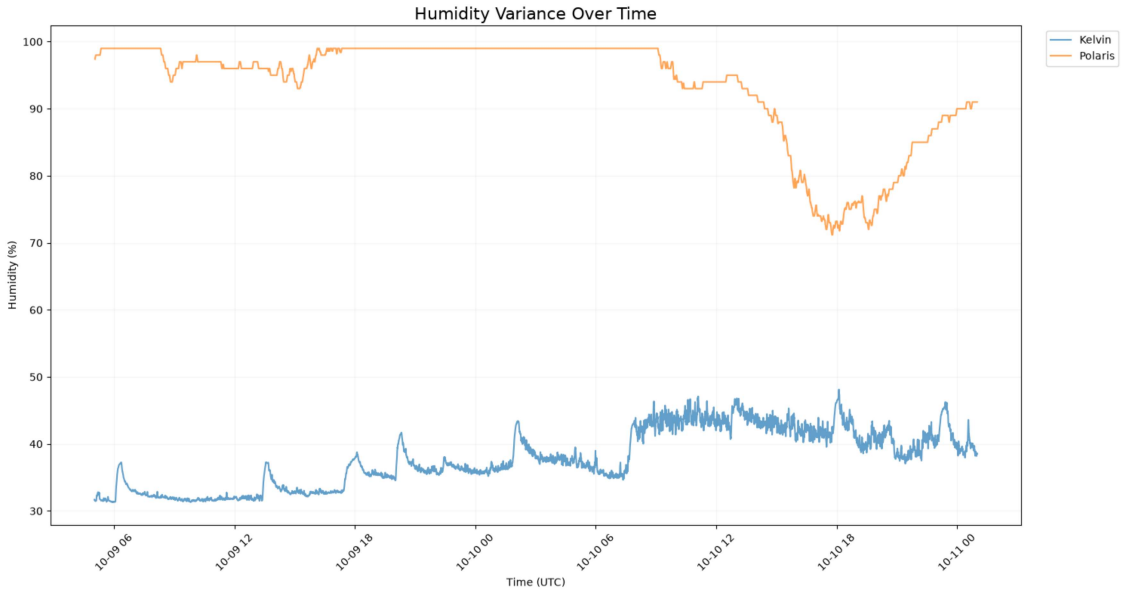
Note: The walls are color coded based on their vulnerability to the wind. Green means sufficient wind breaks or sealants to withstand wind exposure, yellow means there are no tree windbreaks nearby and soffits, porches, or other lift points are present for the wind to catch onto. Red implies a known area of intrusion when the wind blows towards that wall.

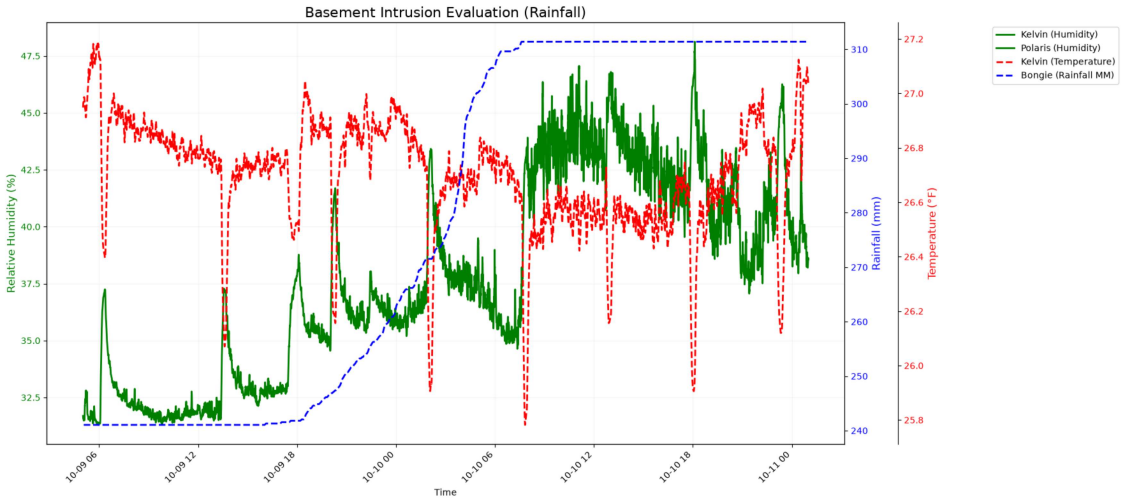


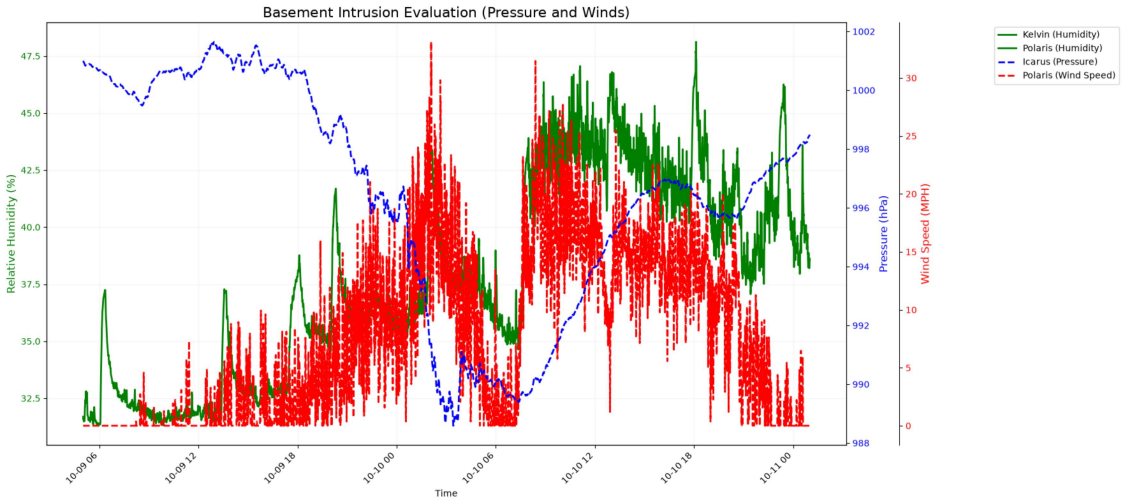
2 Basement Intrusion Post-Mortem

The basement space has been a primary concern since the project started; however this event highlighted the importance of studying that space. On 10/4/2026 around 16:00, water began impacting the air relative humidity faster than the dehumidifiers could extract the water.

Note: Manual initiative to dry up the puddles occurred around the west wall in the basement around 22:00:00 which reduced the amount of water evaporating into the basement air.







Note: Kelvin is stationed in the basement, Polaris on the rooftop, Icarus in the patio within the shade, and Bongie is the patio rain gauge station.

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3 Overall Envelope Impact Assessment

- **Wind Exposure:** Maximum recorded sustained wind speed: 51.5 mph. (10-10-2026, Polaris)
  - Assessment: Moderate wind, minor envelope damage. Overall the envelope, and foliage on the property withstood the event.
- **Envelope Damage:** Significant water intrusion from the basement walls and in the crawlspace. Recommended investigation into potential solutions for preventing water intrusion through the west wall of the Basement and through the vents of the crawlspace. This

can be linked directly to wind blowing rain into exterior openings (the crawlspace and the fireplace) and the ground water table (amount of rain + wind direction)
  - Significant water intrusion via the west and north walls of the basement. (Primarily the west wall from ground saturation.)
  - Significant water intrusion via the fireplace. (Windblown rain?)
  - Significant water intrusion via the west wall of the crawlspace. (Windblown rain through vents.)
- **Water Impacts:**
  - Very noticeable water intrusion in the basement seeping through walls. Puddling near equipment and plastic totes. Potential causes include heightened water table near the west wall causing hydrostatic pressure. This causes the water to push through the wall due to the pressure. Ground slopes toward and around the west wall and foundation of the envelope.
  - Significant humidity anomalies detected in the basement area (Elevated from 30% baseline to sustained nearly 50% during rainfall events and when water intrusion was at its peak. These anomalies were very similar to the last low pressure systems **Tropical Storm Arthur** and **LPS002**)

4 Equipment and Accuracy Assessment

The stations overall handled the event very well with minor disturbances in the outside environment. Internet outage caused the remaining first generation stations to significantly drift away from the second generation fleet in one case. (The other was slightly drifted.

In contrast to LPS002 (General low pressure event last weekend), this system was tracked with clock drift on the first generation stations as a result of the internet outage. Second generation stations outperformed the first generation in every tracked metric (with a much higher resolution).

4.1 Architects Statement

> "Several features of the envelope were tested and compromized. We will be taking active measures in mitigating the risks exposed by this event."
>
> ~ Charles Harding - System Architect and Building Scientist.

4.1.1 Disclaimer

This report was generated based on data obtained by The Regulus Project in 2026. It is solely for scientific research purposes and is not an authoritative truth. The data within this report are the observations reported by the station mesh and may not represent the full resolution of the observations.

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